

Customer Support Analytics

From 8,469 raw tickets to a validated SQL Server star schema and a five-page Power BI decision-support report.

8,469

support tickets

67.30%

backlog rate

35

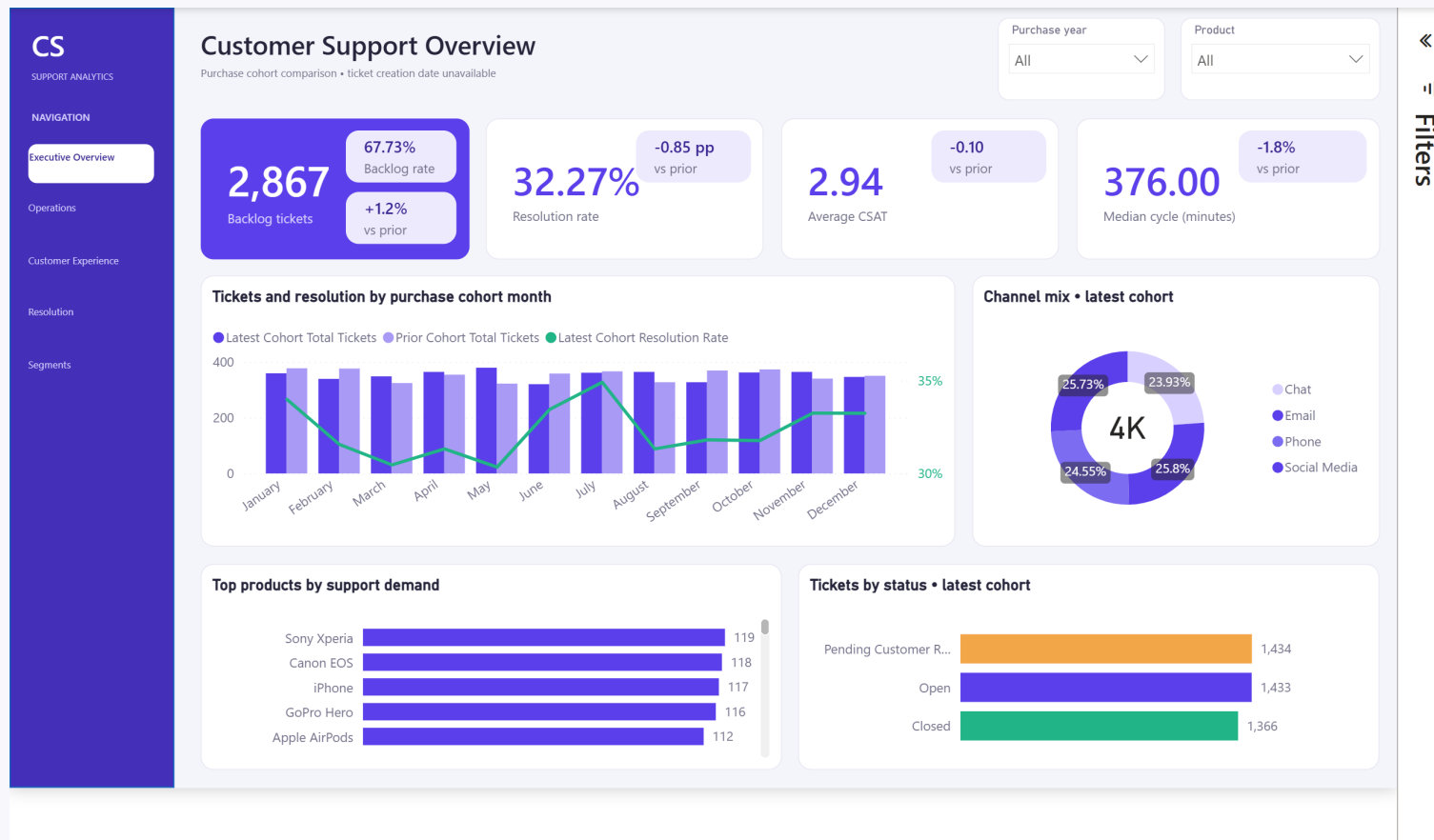
DAX measures

21 / 21

Python tests

5

report pages



Python | pandas | SQL Server | Dimensional Modeling | DAX | Power BI | Git

Business problem and delivery architecture

A simulated Customer Experience team needs a reliable view of unresolved workload, service demand, satisfaction, resolution performance, and customer segments. The solution separates data engineering from analytical modeling so that quality failures remain auditable.

BRONZE	SILVER	GOLD	SQL	POWER BI
Immutable local Kaggle CSV	Python cleaning, flags, pseudonymization	FactTicket plus seven dimensions	Staging, typed analytics, 34 checks	35 DAX measures and five pages

Analytical model

Component	Design	Reason
FactTicket	One row per validated ticket; 21 fields	Stable business grain and additive ticket counts
7 dimensions	Customer profile, product, issue, channel, priority, status, purchase date	Reusable filtering and controlled dimension grain
7 relationships	Many-to-one, fact-to-dimension	Predictable filter propagation and simpler DAX
_Measures	35 explicit measures in business folders	Governed definitions and consistent formatting
Privacy	Names and emails excluded from public model	Portfolio-safe demographic analysis

Why a star schema?

It makes ticket grain, filter propagation, dimension uniqueness, and foreign-key reconciliation explicit.

Why Import mode?

The portfolio dataset is static and small enough for fast interactive analysis without DirectQuery complexity.

Why purchase cohorts?

Purchase date is available, but ticket-created date is not. The report avoids presenting a false ticket-arrival trend.

Findings with analytical integrity

The dashboard highlights operational risk while making denominator, sample, and source constraints visible.

Workload

5,700 tickets are unresolved; the backlog rate is 67.30%. Critical/High backlog contains 783 tickets.

Customer experience

2,769 tickets have CSAT. Average CSAT is 2.99 and 39.80% of respondents rated service 1 or 2.

Resolution cycle

1,404 records have a valid first-response-to-resolution cycle. Median is 380.50 minutes; average is 454.68.

Channel

Email has the highest observed volume at 2,143 tickets. Chat has the highest observed average CSAT at 3.08.

Issue demand

Refund Request / Hardware issue is the largest observed issue combination with 129 tickets.

Product context

Canon EOS has the highest observed ticket volume at 240, but sales volume is unavailable, so this is not a defect rate.

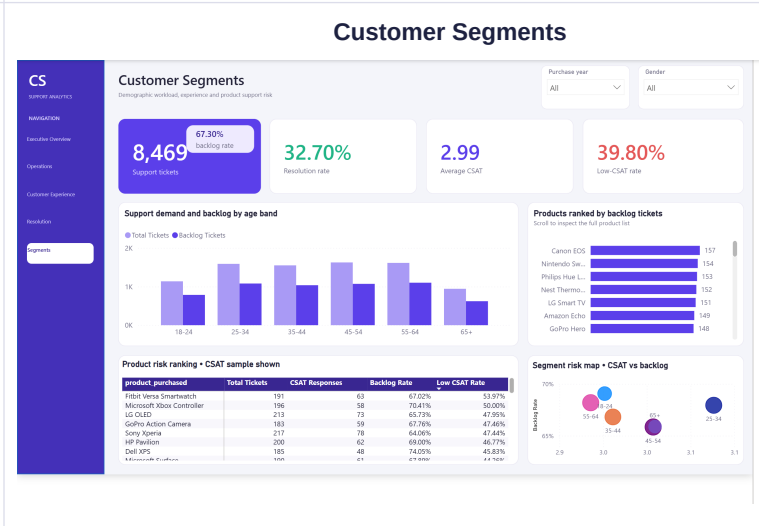
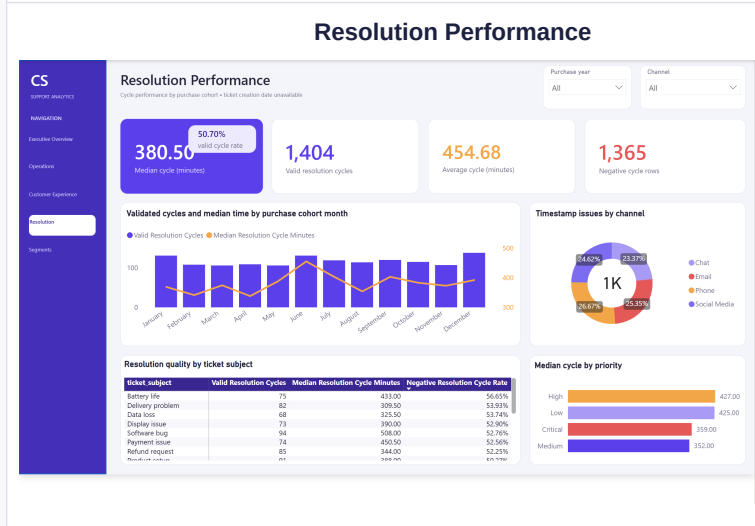
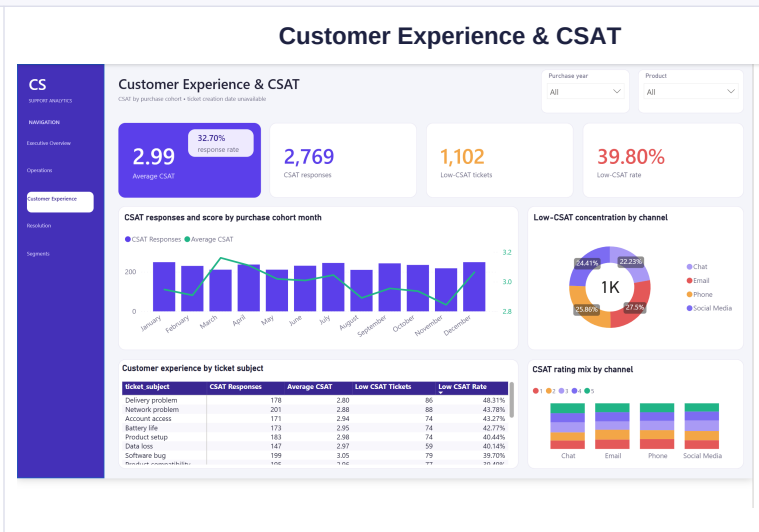
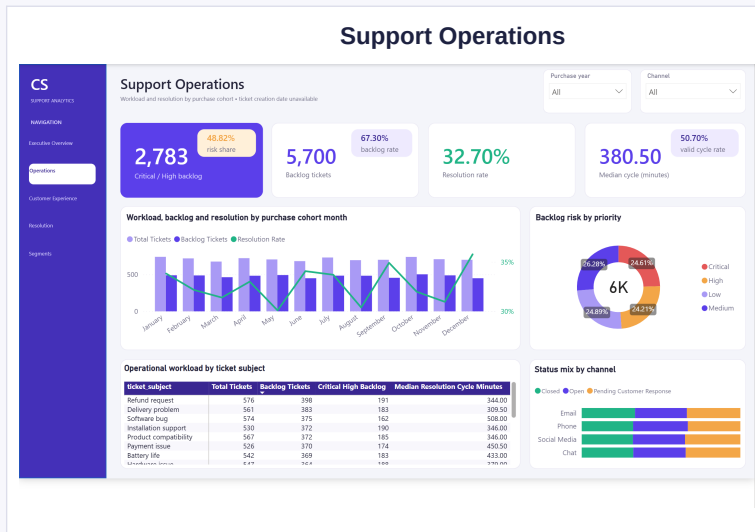
Quality evidence

Check	Result	Interpretation
Raw / Silver / Fact rows	8,469 / 8,469 / 8,469	Complete row reconciliation
Duplicate ticket IDs	0	Fact grain preserved
Missing Gold foreign keys	0	All relationships resolve
Invalid parsed values	0	Age, CSAT, date, and timestamp parsing passed
Negative temporal cycles	1,365 (16.12%)	Flagged and excluded from duration KPIs
Automated validation	21 Python tests + 34 SQL checks	Repeatable quality gates

A negative duration is evidence of a source-quality problem, not a number that should be repaired with `abs()`. The project preserves a quality flag and leaves the analytical duration blank.

Power BI report gallery

Five pages separate executive monitoring, operations, customer experience, resolution quality, and customer segmentation. The Executive Overview appears on the cover; the remaining pages are shown below.



UX controls: consistent page navigation, labeled slicers, sample-size context, explanatory caveats, visual alt text, and deterministic tab order.

Recommendations, limitations, and outcome

1. Prioritize urgency

Review the 783 Critical/High unresolved tickets and monitor their share of backlog.

2. Improve survey coverage

A 32.70% response rate limits how confidently CSAT findings generalize to all tickets.

3. Investigate segments

Use subject, channel, and product rates to select cases for qualitative review, not to claim causation.

4. Repair event capture

Add ticket-created timestamps and prevent resolution events from preceding first response.

5. Add denominators

Capture sales volume, SLA targets, agent capacity, order value, and support cost.

Known limitations

- No explicit ticket-created timestamp.
- Purchase date is not ticket-arrival date.
- CSAT covers respondents only.
- Resolution cycle starts at first response.
- No agent, SLA, sales, revenue, or cost fields.
- Segment comparisons are descriptive, not causal.

Portfolio evidence

- Source-controlled PBIP definitions.
- 21 Python tests and 34 SQL checks.
- Versioned Silver and Gold quality summaries.
- 35 explicit DAX measures.
- Five verified report screenshots.
- Private Power BI Service deployment.

Outcome: a reproducible analytical product that combines engineering, modeling, validation, dashboard design, deployment, and transparent communication of uncertainty.

Repository

<https://github.com/thaquan/CUSTOMER-SUPPORT-TICKET>

Dataset

Customer Support Ticket Dataset by Suraj on Kaggle, CC0 - Public Domain.